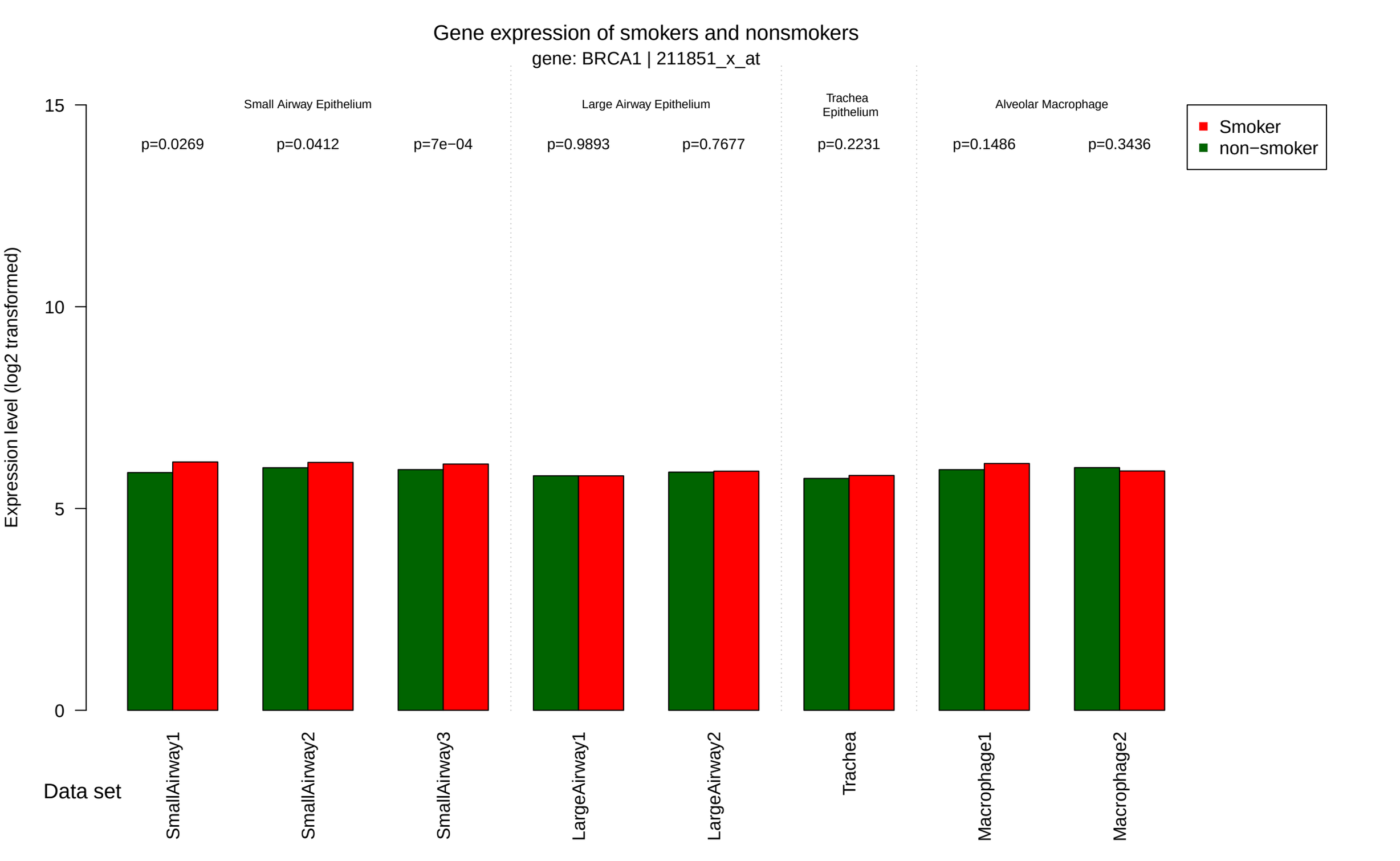
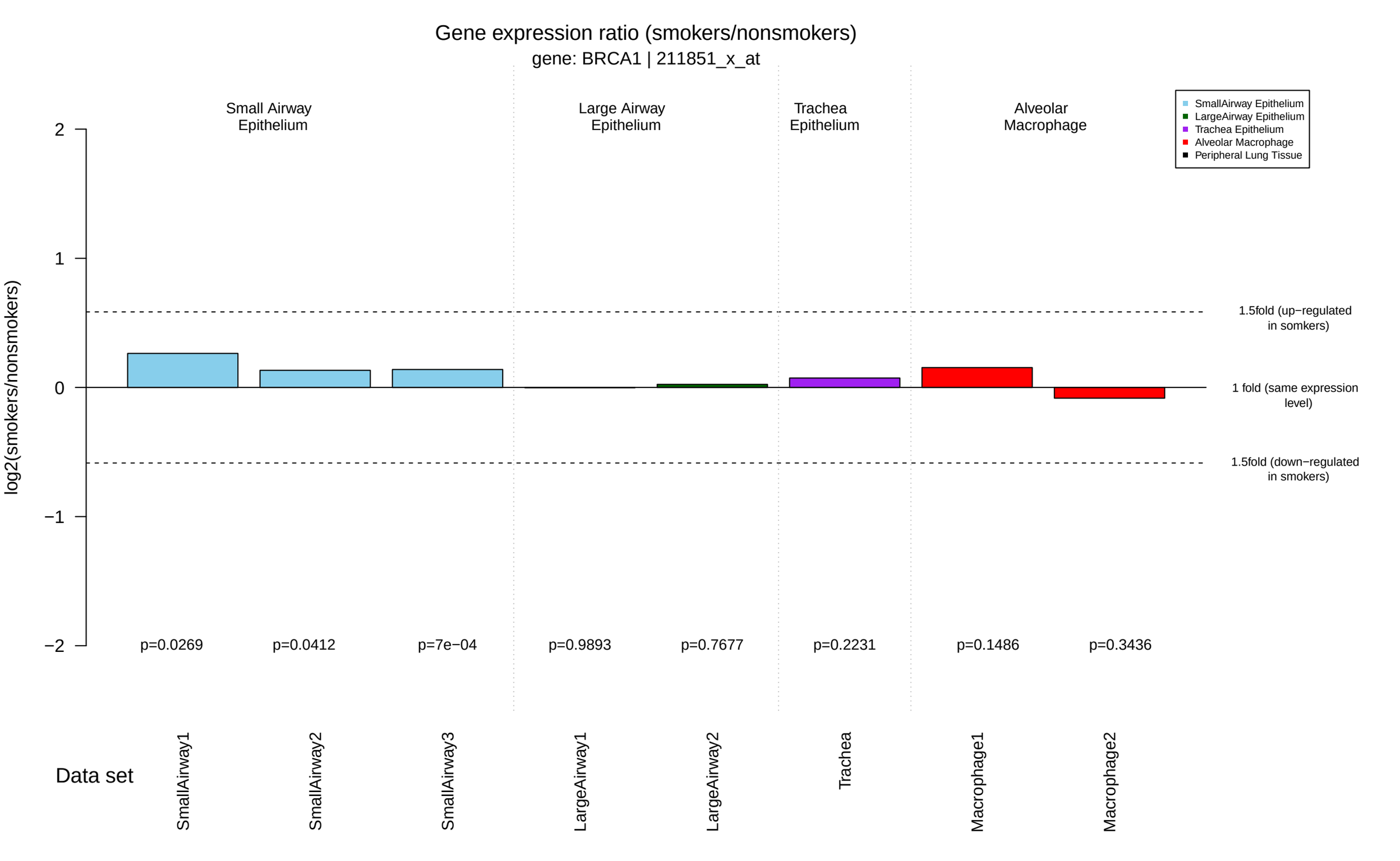
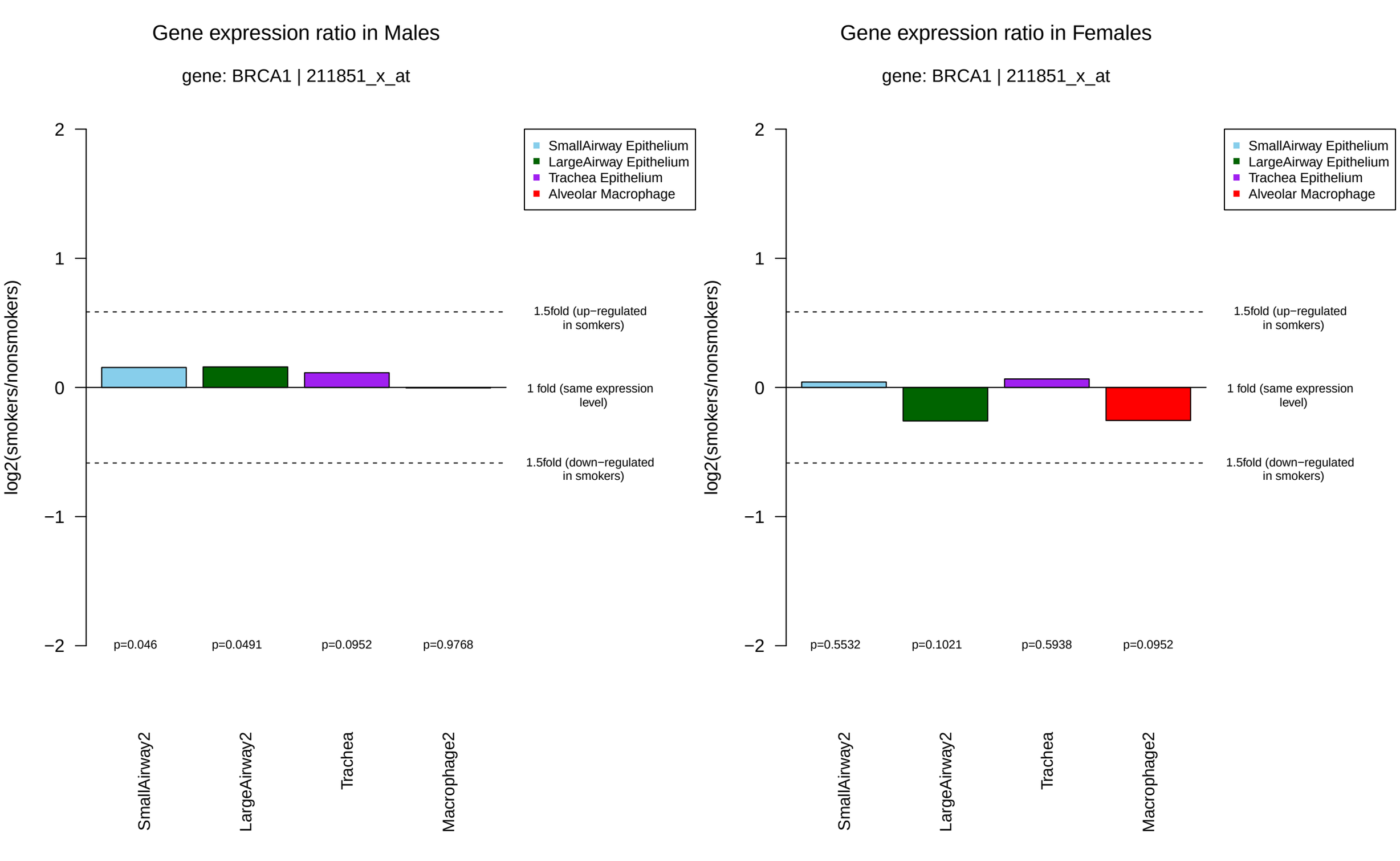
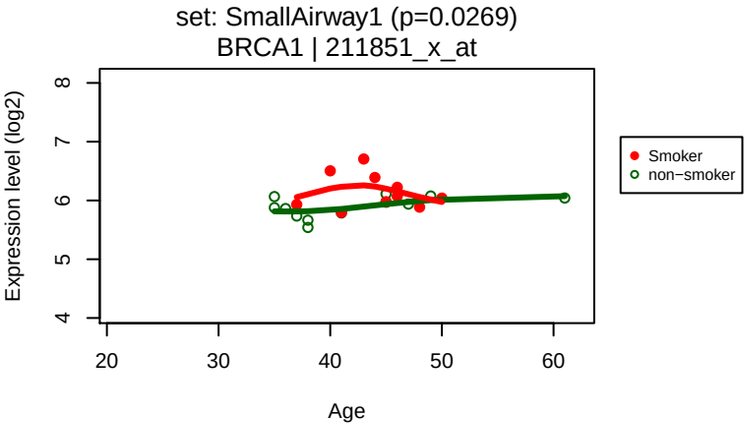
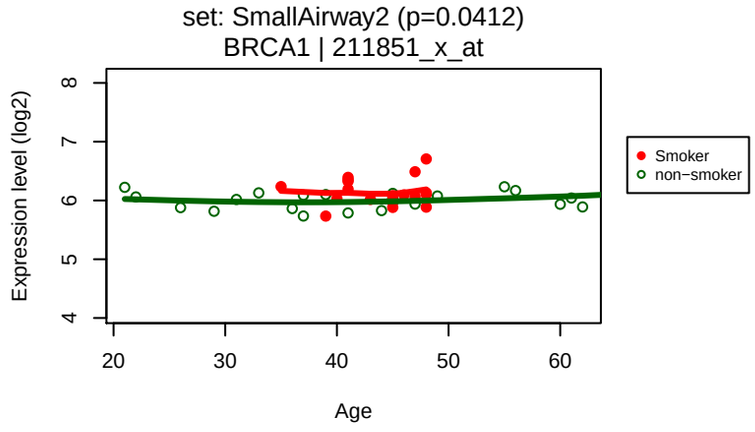
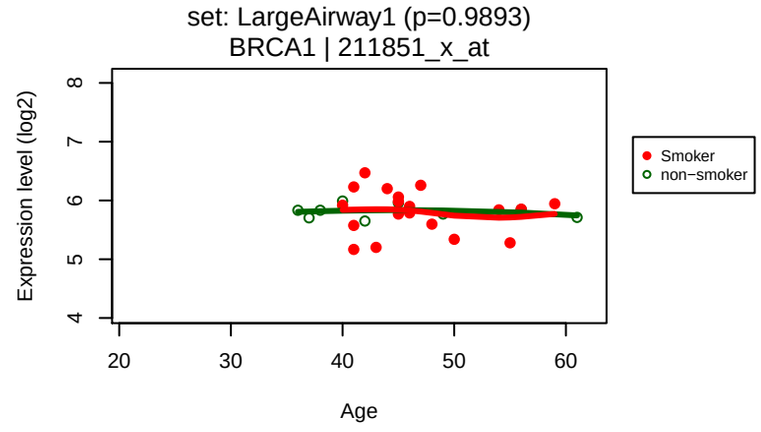

Gene expression of smokers and nonsmokers
gene: BRCA1 | 211851_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0269
p=0.0412
p=7e−04
p=0.9893
p=0.7677
p=0.2231
p=0.1486
p=0.3436
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

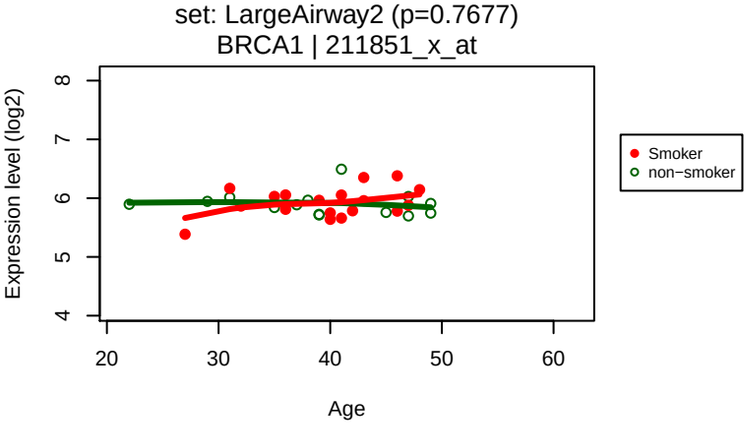

Gene expression ratio (smokers/nonsmokers)
gene: BRCA1 | 211851_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0269
p=0.0412
p=7e−04
p=0.9893
p=0.7677
p=0.2231
p=0.1486
p=0.3436
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

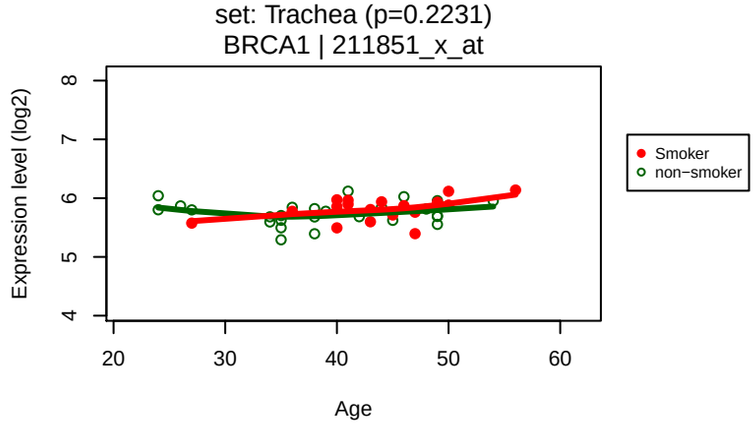

Gene expression ratio in Males
gene: BRCA1 | 211851_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.046
p=0.0491
p=0.0952
p=0.9768
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BRCA1 | 211851_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5532
p=0.1021
p=0.5938
p=0.0952
SmallAirway2
LargeAirway2
Trachea
Macrophage2

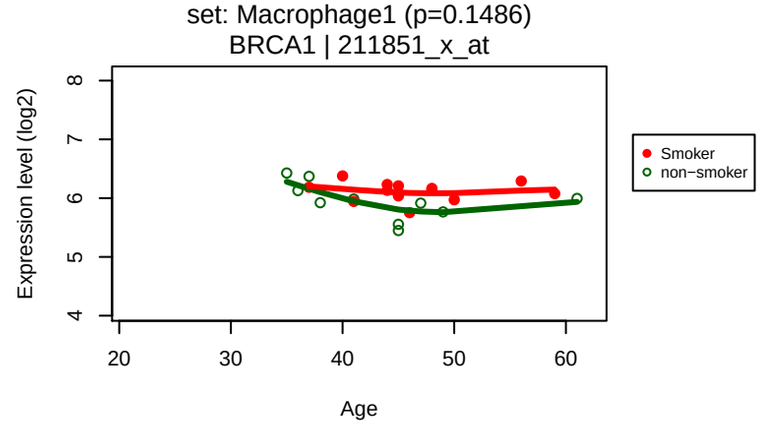

set: SmallAirway1 (p=0.0269)
BRCA1 | 211851_x_at
Expression level (log2)
Age
Smoker
non−smoker

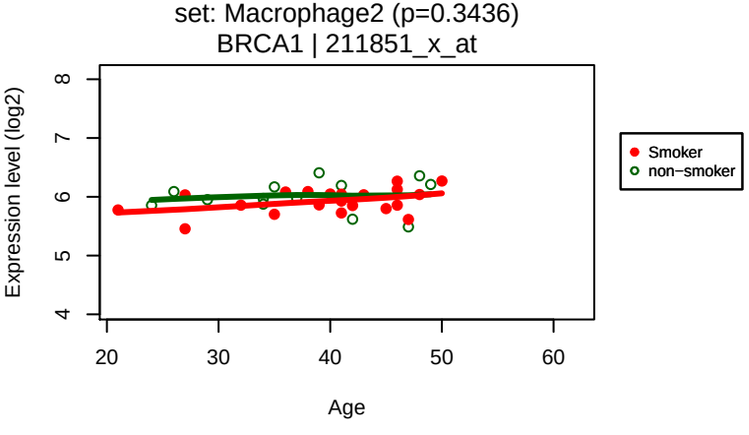

set: SmallAirway2 (p=0.0412)
BRCA1 | 211851_x_at
Expression level (log2)
Age
Smoker
non−smoker

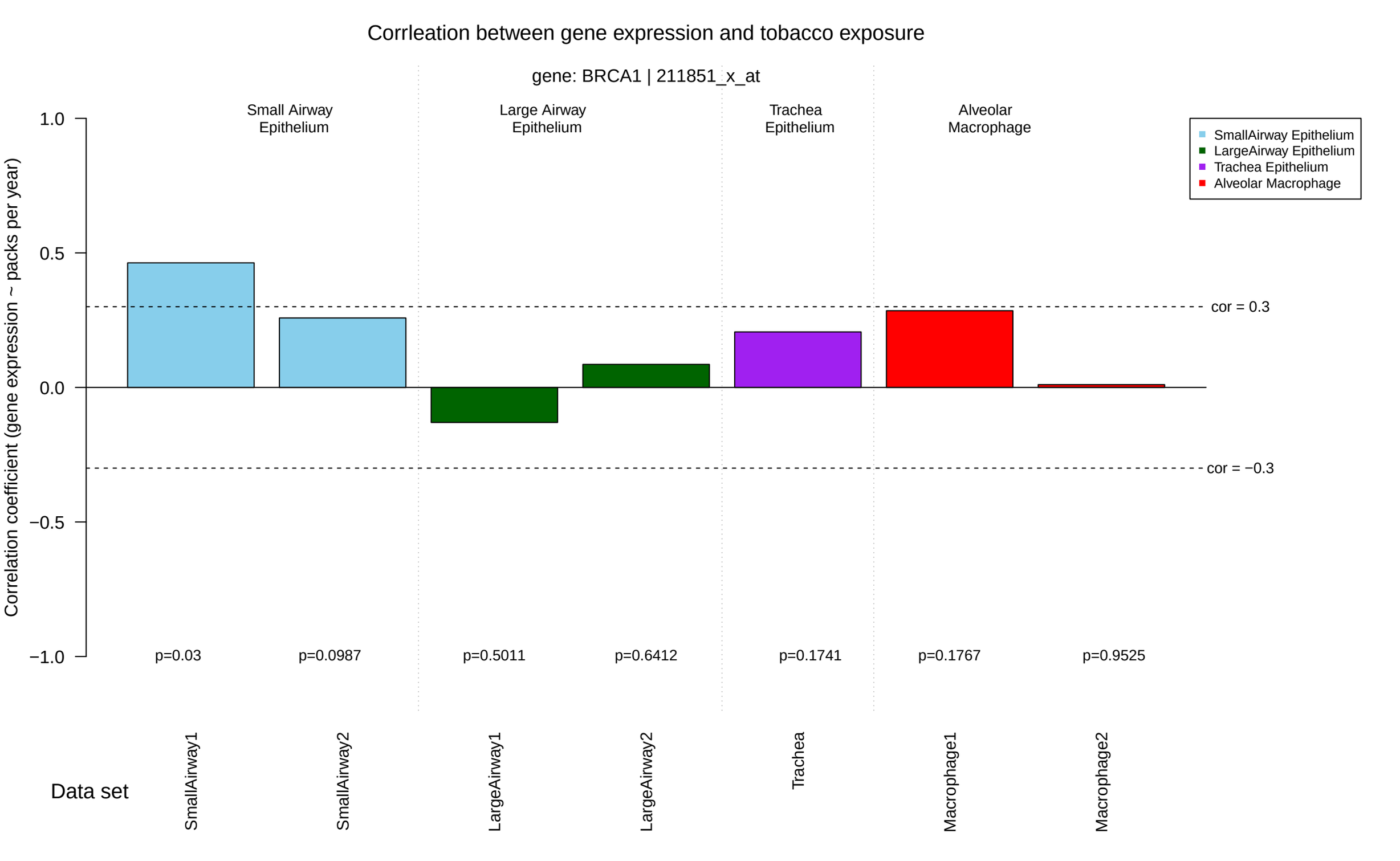

set: LargeAirway1 (p=0.9893)
BRCA1 | 211851_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7677)
BRCA1 | 211851_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2231)
BRCA1 | 211851_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1486)
BRCA1 | 211851_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3436)
BRCA1 | 211851_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BRCA1 | 211851_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.03
p=0.0987
p=0.5011
p=0.6412
p=0.1741
p=0.1767
p=0.9525
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2